

Sunburn device: the whole program

every block of the program that runs on the wrist, explained in plain English · printed 2026-09-15

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights and two buttons. It measures the sun's UV with a sensor, checks the online UV index for where you are through a phone, and then flashes, beeps or taps your arm until you put sunscreen on. Two hours later it reminds you to reapply. The program that does all this is built from 76 coloured blocks. Every one of them is on these sheets, with a note saying what it does.

Each picture is one block of the program, exactly as it looks in the MakeCode editor where the program is built by snapping coloured pieces together. A block is read from top to bottom. Under each picture: the block's name, what kind of block it is, what it does, which other blocks use it (Used by) and which blocks it calls on (Uses). This sheet is the whole program at a glance: nine numbered sections. Sections 1 and 2 are the plan the device follows; sections 3 to 9 are the parts that plan calls on.



- 1

Start here: settings · 1 block · sheet 2
This block runs first, the moment the device is switched on. It holds every setting: which UV sensor is fitted, how the sound and the tapping arm are wired, the UV levels that count as low, moderate, very high and extreme, and how long each wait is. Then it starts everything up. To change how the device behaves, this is the only place to look.
- 4

Alerts · 4 blocks · sheet 3
The ways the device gets your attention: flashing and beeping, tapping your arm with the small motor, and going to sleep when nobody answers. Each one keeps going until you press A or its time limit runs out. No answer for a long time means the device is probably not being worn, so it sleeps until a button is pressed.
- 7

Talking to the phone · 12 blocks · sheet 4
How the wearable and the phone app talk over Bluetooth. The phone sends the online UV index for where you are; the wearable sends back what it is doing every two seconds and reports each event, such as sunscreen applied. The same messages also go down the USB cable, so everything can be tested on a laptop. Every message ends with a checksum, so a message damaged on the way is thrown away instead of being misread.

- 2

The flowchart · 20 blocks · sheet 2
The plan the device follows all day, taken straight from the Sunburn Flowchart. Every box and every question on the flowchart is one block here, with the same name, in the same order: read the UV sensor, check the reading makes sense, ask the phone for the online UV index, decide how strong the sun is, then react. The forever block keeps the plan running round and round.
- 5

Screen and lights · 8 blocks · sheet 3
What you see. The 5 by 5 grid of lights shows the UV number, a happy face or a warning, and messages one word at a time. If a small screen is fitted, it shows the number large, the level, the message and the countdowns.
- 8

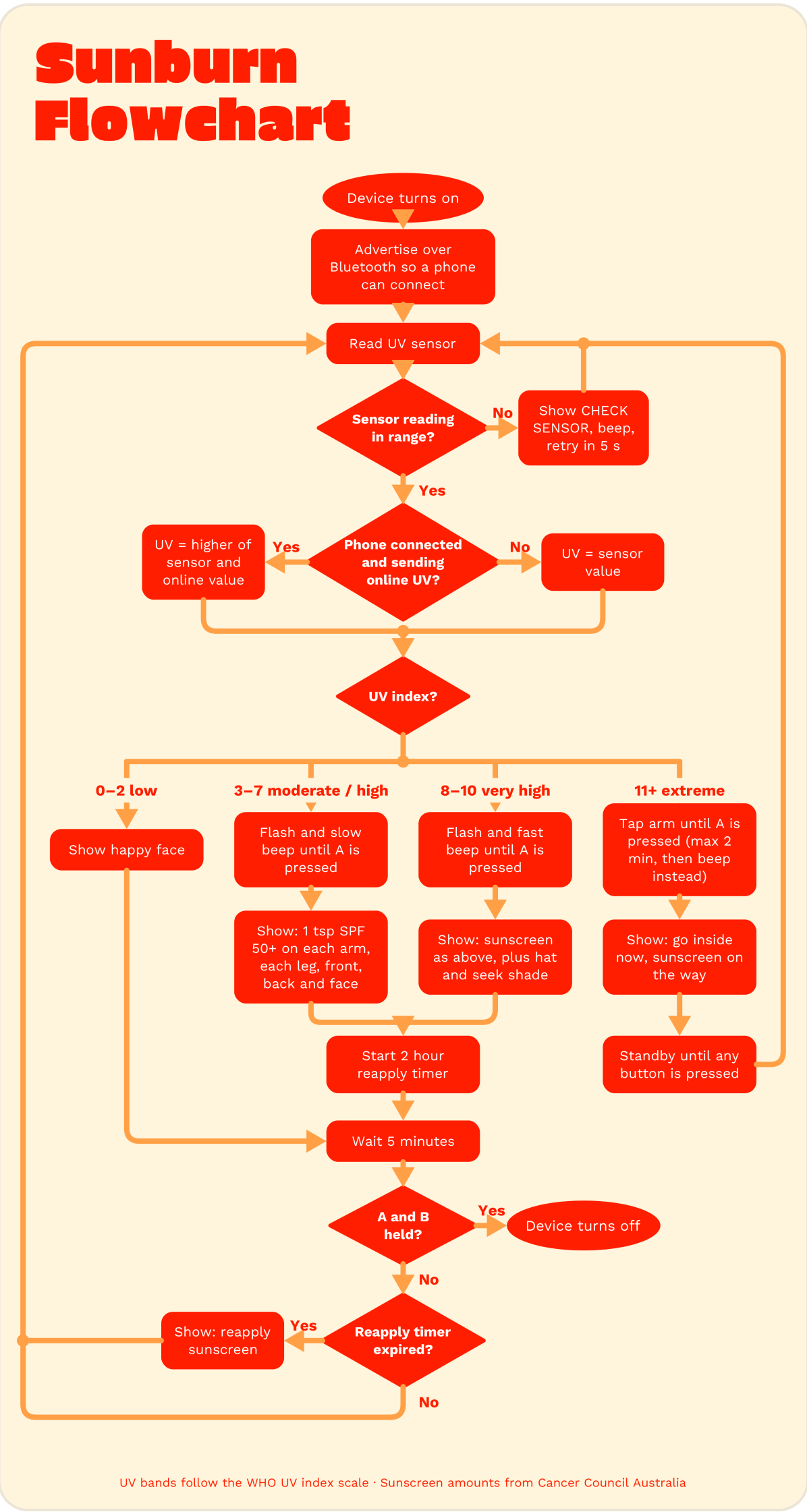
Helpers · 11 blocks · sheet 5
Small jobs the other blocks share: the maths for each type of sensor, turning minutes into the milliseconds the program counts in, sorting numbers, tidying up numbers, times and text, and the checksum that goes on the end of every message.

- 3

Buttons · 3 blocks · sheet 3
The micro:bit has two buttons. A means done: sunscreen is on, or I am going inside. B wakes the device up. Pressing both together turns it off, or on again.
- 6

Sound and the tapping arm · 5 blocks · sheet 4
The beeps, and the small motor (a servo) whose arm taps your wrist when the sun is extreme.
- 9

The optional screen · 12 blocks · sheet 6
Only used if a small screen is added to the wearable. These blocks draw letters and numbers on it dot by dot. The device works exactly the same without a screen.



The Sunburn Flowchart the device follows. Every box on it is a block in section 2 with the same name.

- Variables: remember a value

Text
- Functions: a step with a name

Arrays: lists
- Basic: the lights, pauses

Music: beeps
- Input: buttons and time

Pins: sensor, motor, screen
- Loops: repeat

Bluetooth: the phone
- Logic: if, compare

Serial: the USB cable
- Maths

Control: timing

Reading the other sheets

Every picture is one block from the editor at 60% of its size on screen. Under it: the block's name, what kind of block it is, what it does, which blocks use it (Used by) and which blocks it calls on (Uses).

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

1

Start here: settings

This block runs first, the moment the device is switched on. It holds every setting: which UV sensor is fitted, how the sound and the tapping arm are wired, the UV levels that count as low, moderate, very high and extreme, and how long each wait is. Then it starts everything up. To change how the device behaves, this is the only place to look.

1 block

on start

set uvSensorType to 1

set uvSampleCount to 15

set soundOutput to 1

set soundVolume to 180

set servoType to 1

set servoRestAngle to 90

set servoTapAngle to 30

set oledMode to 1

set oledType to 1

set oledAddress to 60

set uvLowMax to 2

set uvHighMax to 7

set uvVeryHighMax to 10

set waitMinutes to 5

set reapplyHours to 2

set tapMaxMinutes to 2

set alertGiveUpMinutes to 10

set onlineUvMaxAgeMinutes to 30

set sensorRetrySeconds to 5

set usbSpeed to 1

set firmwareVersion to 3.2

call applySensorPreset

call applyTimings

call setupSound

call setupOled

call setupUsbLink

call advertiseOverBluetooth

if button B is pressed then

set demoMode to true

call applyTimings

call deviceTurnsOn

on start Runs once, when the device is switched on. The first thing that runs when the device is switched on. All the settings are listed here, then the start-up steps: work out the sensor maths, set the timings, get the sound, the screen and Bluetooth ready, and show that the device is on. Holding B while switching on uses the fast demo timings.

Uses: deviceTurnsOn, setupSound, setupUsbLink, advertiseOverBluetooth, applySensorPreset, applyTimings, setupOled

2

The flowchart

The plan the device follows all day, taken straight from the Sunburn Flowchart. Every box and every question on the flowchart is one block here, with the same name, in the same order: read the UV sensor, check the reading makes sense, ask the phone for the online UV index, decide how strong the sun is, then react. The forever block keeps the plan running round and round.

20 blocks

forever

if deviceOn then

call runFlowchart

else

pause (ms) 500

forever Runs again and again, all day. Runs the flowchart once, then goes straight back to the top and runs it again, all day long.

Uses: runFlowchart

function runFlowchart

if call sensorReadingInRange then

call chooseUvValue

call showUvReading

set currentBand to call uvBand currentUv

set wentToStandby to false

if currentBand == low then

call showHappyFace

else if currentBand == modhigh then

call moderateOrHighUv

else if currentBand == vhigh then

call veryHighUv

else

call extremeUv

if deviceOn and not wentToStandby then

call wait5Minutes

if deviceOn and call reapplyTimerExpired then

call showReapplySunscreen

else

call showCheckSensor

runFlowchart A step of the program. The whole flowchart from top to bottom. Read the sensor. If the reading makes sense, pick the UV value, show it, decide the level and react: a happy face, or an alert until A is pressed. Then wait 5 minutes and check whether it is time for more sunscreen. If the reading does not make sense, show CHECK SENSOR instead. Pressing A and B together at any moment turns the device off and stops every wait.

Used by: forever · **Uses:** readUvSensor, sensorReadingInRange, showCheckSensor, chooseUvValue, uvBand, showHappyFace, moderateOrHighUv, veryHighUv, extremeUv, wait5Minutes, reapplyTimerExpired, showReapplySunscreen, showUvReading

function readUvSensor

set stateName to reading

repeat uvSampleCount times

do

samples add value analog read pin analog pin P1 to end

pause (ms) 2

call sortNumbers samples

set sensorRaw to samples get value at floor uvSampleCount / 2

set sensorNoise to samples get value at uvSampleCount - 1 samples get value at 0

set sensorVolts to sensorRaw * 3.3 / 1023

set sensorUv to sensorVolts - uvZeroVolts / uvVoltsPerIndex

call debugLog join raw sensorRaw noise sensorNoise

readUvSensor A step of the program. Reads the UV sensor plugged into pin P1. It takes 15 quick readings and uses the middle one, so a single odd reading cannot set off a false alarm. Then it turns the sensor's voltage into a UV index number.

Used by: runFlowchart, handleCommand, calibrateToKnownUv · **Uses:** sortNumbers, debugLog

function showCheckSensor

set stateName to error

if not sensorErrorReported then

call sendLine ev=error;msg=CHECK SENSOR

set sensorErrorReported to true

set sensorErrorCount to 0

show icon

if remainder of sensorErrorCount / 12 == 0 then

call showMessage CHECK SENSOR

call beep 440 200

call beep 440 200

change sensorErrorCount by 1

pause (ms) sensorRetrySeconds * 1000

showCheckSensor A step of the program. When the reading is not believable: shows a cross, and the first time scrolls CHECK SENSOR and beeps twice, then again about once a minute while the fault lasts. It tells the app once, and tries again every 5 seconds. A faulty sensor is never treated as safe.

Used by: runFlowchart · **Uses:** showMessage, oledFooter, beep, sendLine, formatNumber

function extremeUv

set stateName to alert

call tapArmUntilA

if deviceOn then

call showMessage Go inside now. Sunscreen on the way

if ackPressed then

call sendEvent inside

call standbyUntilButton

extremeUv A step of the program. UV 11 or more: taps the arm until A is pressed (after 2 minutes it beeps instead), shows Go inside now, then sleeps until a button is pressed.

Used by: runFlowchart · **Uses:** tapArmUntilA, standbyUntilButton, showMessage, sendEvent

function sunscreenApplied level

show icon

set protectionLevel to level

change sunscreenCount by 1

call sendEvent sunscreen

call start2HourReapplyTimer

sunscreenApplied A step of the program · needs: level. A was pressed after a sunscreen alert: remembers which advice was given, counts one more application, tells the app, and starts the timer.

Used by: moderateOrHighUv, veryHighUv · **Uses:** start2HourReapplyTimer, sendEvent

function sensorReadingInRange

set inRange to true

if sensorRaw <= 1020 then

set inRange to false

if sensorNoise > 200 then

set inRange to false

if sensorUv < -1.5 or sensorUv > 20 then

set inRange to false

if inRange then

set sensorErrorReported to false

return inRange

sensorReadingInRange A yes-or-no question the program asks. Asks: is this reading believable? The answer is no if the sensor is missing or shorted, if the readings jump about (a loose wire), or if the number could not be real.

Used by: runFlowchart

function phoneSendingOnlineUv

set age to running time (ms) - onlineUvTime

set maxAge to onlineUvMaxAgeMinutes * 60000

return onlineUvTime > 0 and age < maxAge

phoneSendingOnlineUv A yes-or-no question the program asks. Asks: has the phone app sent an online UV index in the last 30 minutes?

Used by: chooseUvValue, showUvReading, sendStatus

function chooseUvValue

set currentUv to max of 0 and sensorUv

if call phoneSendingOnlineUv then

set currentUv to max of currentUv and onlineUv

chooseUvValue A step of the program. Picks the UV number the device will act on: the higher of the sensor's reading and the phone's online value, or just the sensor's reading when the phone is not sending.

Used by: runFlowchart · **Uses:** phoneSendingOnlineUv

function showHappyFace

set stateName to safe

show icon

call oledShowMessage Low UV. No sunscreen needed

showHappyFace A step of the program. UV 0 to 2: shows a happy face. No sunscreen needed right now.

Used by: runFlowchart · **Uses:** oledShowMessage

function moderateOrHighUv

if protectionLevel == 0 then

set stateName to alert

call showMessage Sunscreen time

call flashAndBeepUntilA false

if ackPressed then

call showMessage 1 tsp SPF 50+ on each arm, each leg, front, back

call sunscreenApplied 1

else if deviceOn then

call standbyUntilButton

else

call alreadyProtected

moderateOrHighUv A step of the program. UV 3 to 7: flashes and beeps slowly until A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If sunscreen is already on, it just shows a tick and the time left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

Used by: runFlowchart · **Uses:** sunscreenApplied, alreadyProtected, flashAndBeepUntilA, standbyUntilButton, showMessage

function veryHighUv

if protectionLevel < 2 then

set stateName to alert

call showMessage Very high UV

call flashAndBeepUntilA true

if ackPressed then

call showMessage 1 tsp SPF 50+ on each arm, each leg, front, back

call sunscreenApplied 2

else if deviceOn then

call standbyUntilButton

else

call alreadyProtected

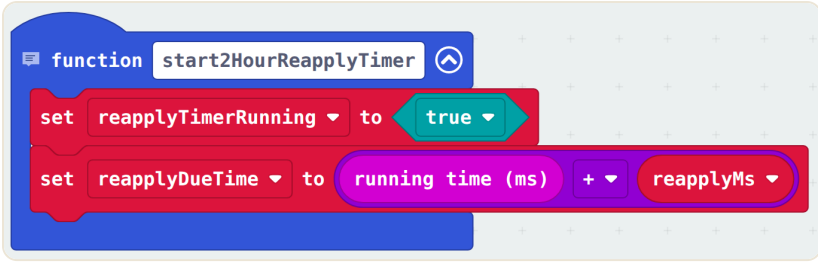
veryHighUv A step of the program. UV 8 to 10: flashes and beeps quickly until A is pressed, then shows the sunscreen advice plus hat and shade, and starts the 2 hour timer. It alerts again if the wearer so far only had the moderate advice.

Used by: runFlowchart · **Uses:** sunscreenApplied, alreadyProtected, flashAndBeepUntilA, standbyUntilButton, showMessage

2 The flowchart (continued)

20 blocks

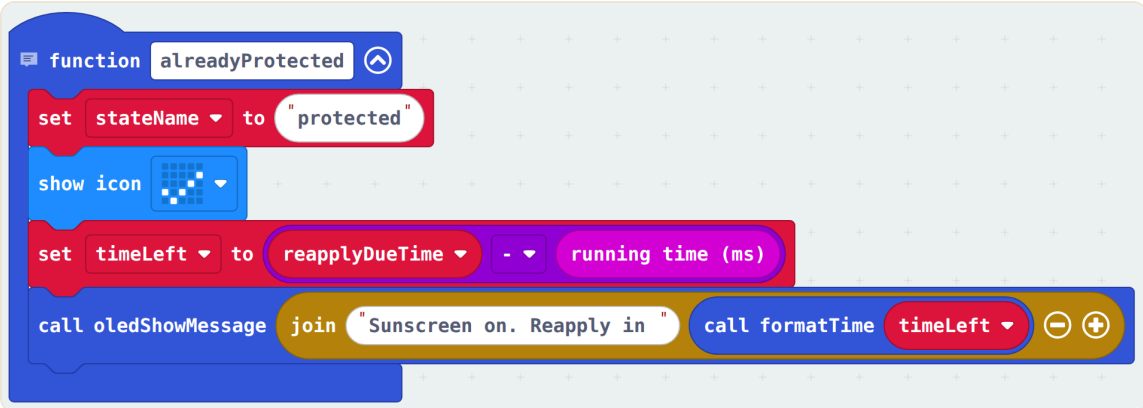
The plan the device follows all day, taken straight from the Sunburn Flowchart. Every box and every question on the flowchart is one block here, with the same name, in the same order: read the UV sensor, check the reading makes sense, ask the phone for the online UV index, decide how strong the sun is, then react. The forever block keeps the plan running round and round.



start2HourReapplyTimer A step of the program

Starts the 2 hour countdown to the next sunscreen.

Used by: sunscreenApplied



alreadyProtected A step of the program

Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

Used by: moderateOrHighUv, veryHighUv · Uses: oledShowMessage, formatTime



reapplyTimerExpired A yes-or-no question the program asks

Asks: have the 2 hours since the last sunscreen passed?

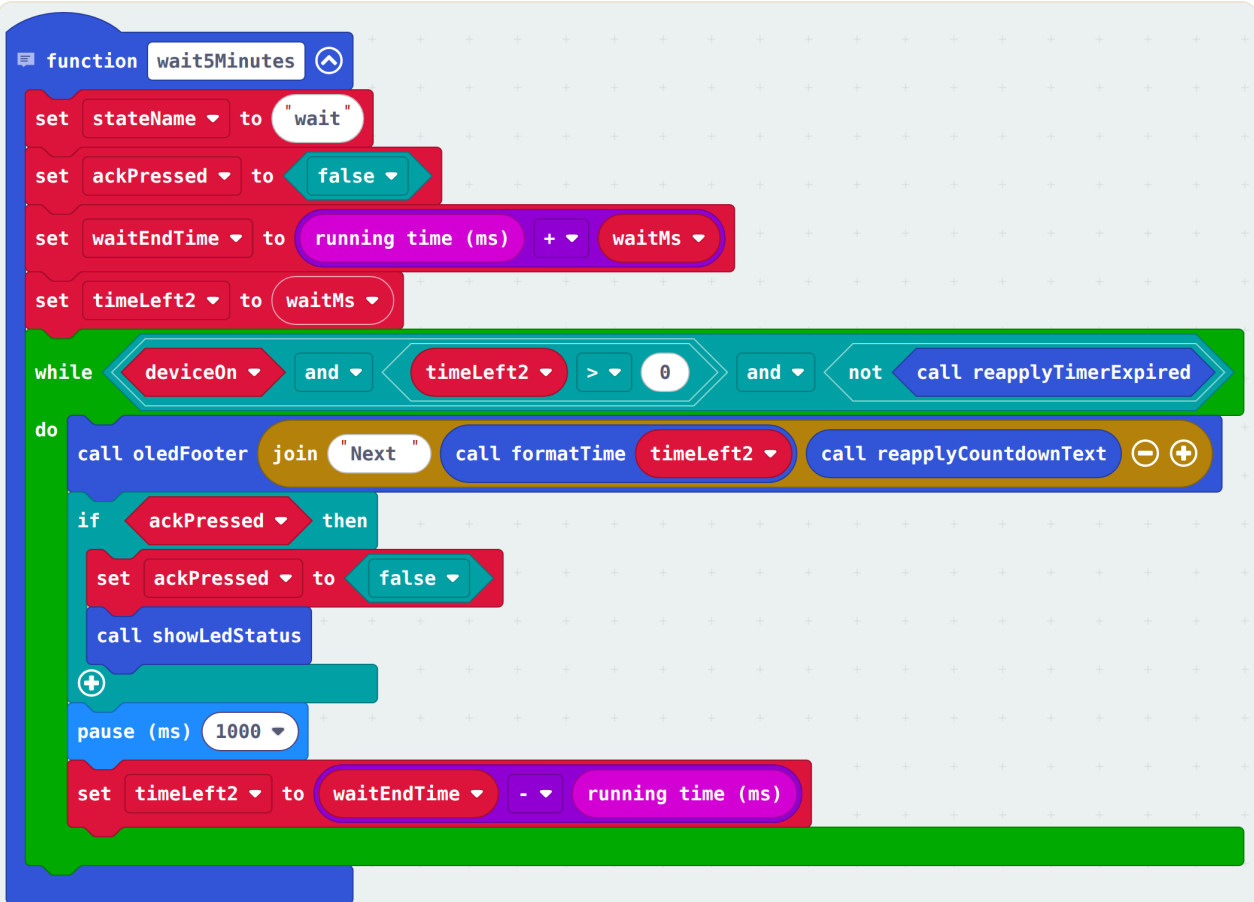
Used by: runFlowchart, wait5Minutes



deviceTurnsOn A step of the program

Switching on: shows a tick, says Device on on the screen, tells the app, and marks the device as on.

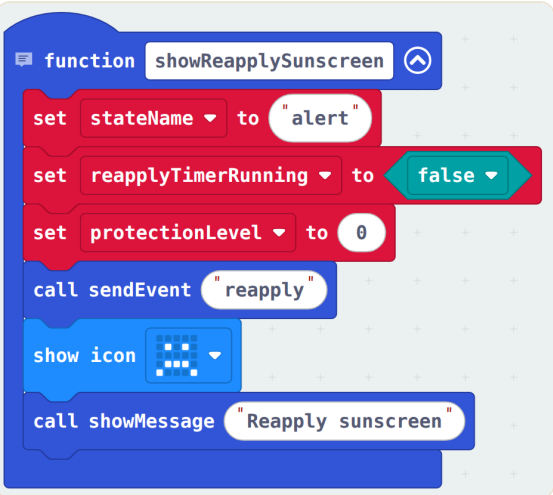
Used by: on button A+B pressed, handleCommand, on start · Uses: oledHeader, oledShowMessage, oledFooter, servoOff, sendEvent



wait5Minutes A step of the program

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

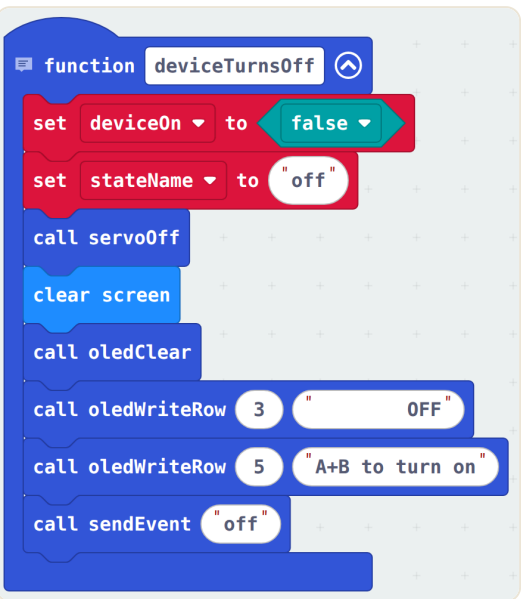
Used by: runFlowchart · Uses: reapplyTimerExpired, showLedStatus, oledFooter, formatTime, reapplyCountdownText



showReapplySunscreen A step of the program

Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.

Used by: runFlowchart · Uses: showMessage, sendEvent



deviceTurnsOff A step of the program

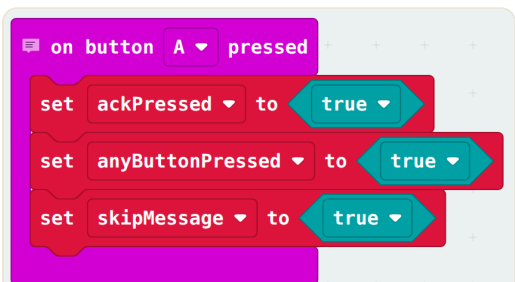
Switching off: everything stops and the screen shows OFF. Bluetooth stays on so the app can switch it back on.

Used by: on button A+B pressed, handleCommand · Uses: servoOff, sendEvent, oledClear, oledWriteRow

3 Buttons

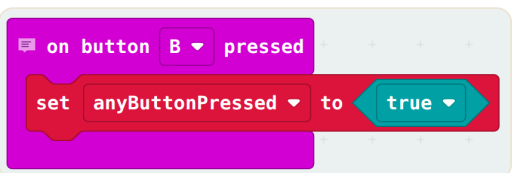
3 blocks

The micro:bit has two buttons. A means done: sunscreen is on, or I am going inside. B wakes the device up. Pressing both together turns it off, or on again.



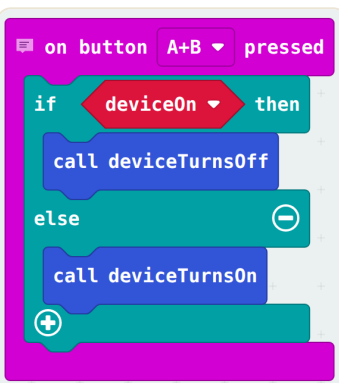
on button A pressed Runs each time button A is pressed

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.



on button B pressed Runs each time button B is pressed

Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.



on button A+B pressed Runs when A and B are pressed together

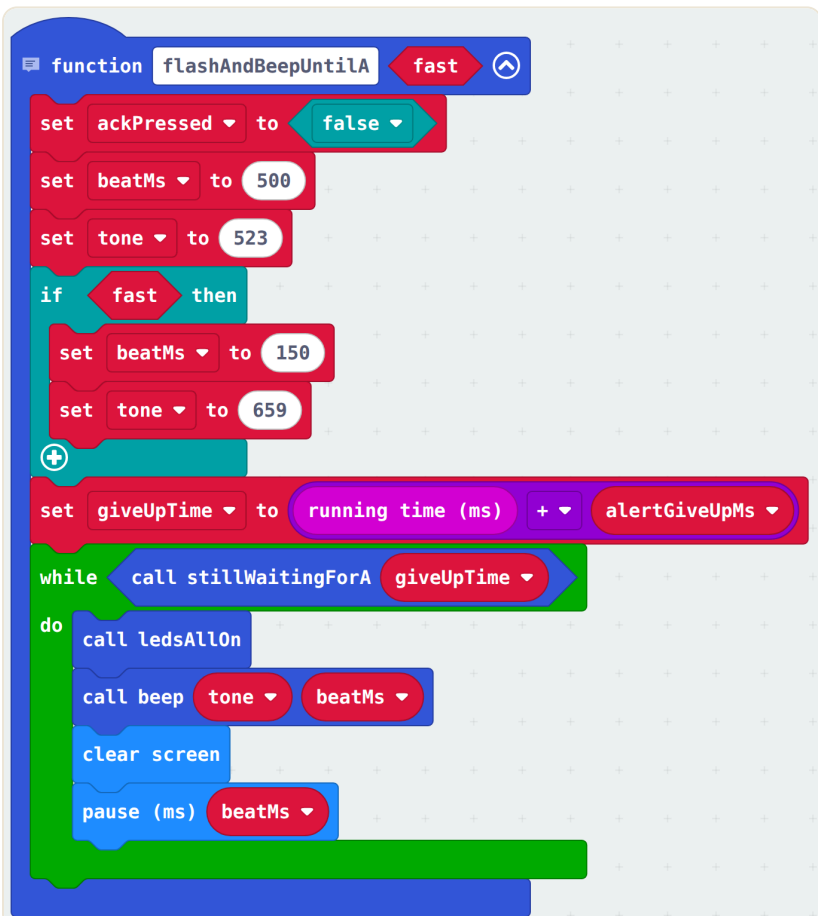
Pressing A and B together turns the device off, or on again.

Uses: deviceTurnsOn, deviceTurnsOff

4 Alerts

4 blocks

The ways the device gets your attention: flashing and beeping, tapping your arm with the small motor, and going to sleep when nobody answers. Each one keeps going until you press A or its time limit runs out. No answer for a long time means the device is probably not being worn, so it sleeps until a button is pressed.



flashAndBeepUntilA A step of the program · needs: fast
Flashes all 25 lights and beeps until A is pressed or the app says done. The fast setting is the quicker, higher beep for very high UV. It gives up after 10 minutes so a device left on a table does not beep all day.

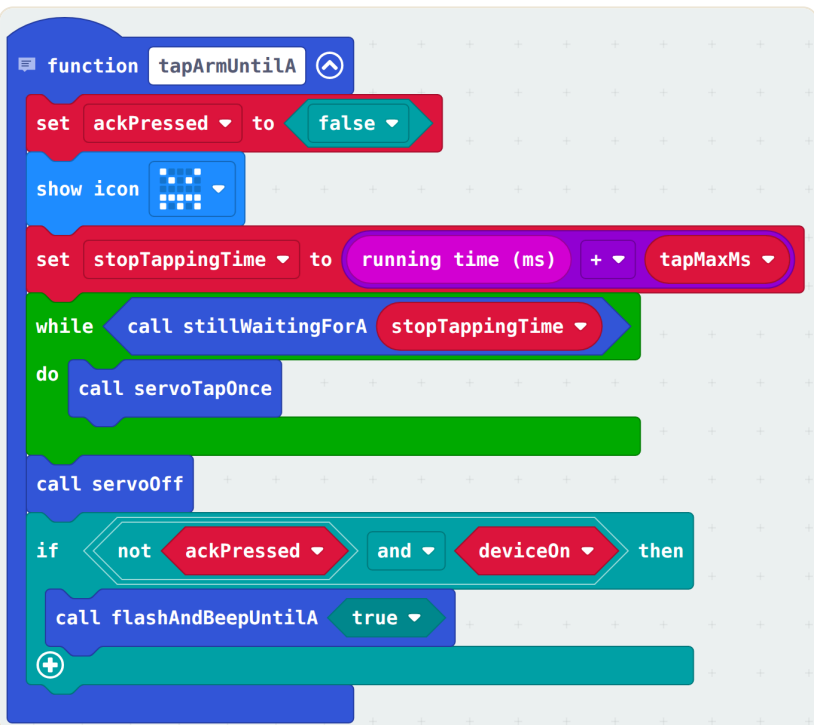
Used by: moderateOrHighUv, veryHighUv, tapArmUntilA · Uses: stillWaitingForA, ledsAllOn, beep



stillWaitingForA A yes-or-no question the program asks · needs: untilTime

Asks: should the alert carry on? Yes while A has not been pressed, the device is still on, and the time limit has not been reached.

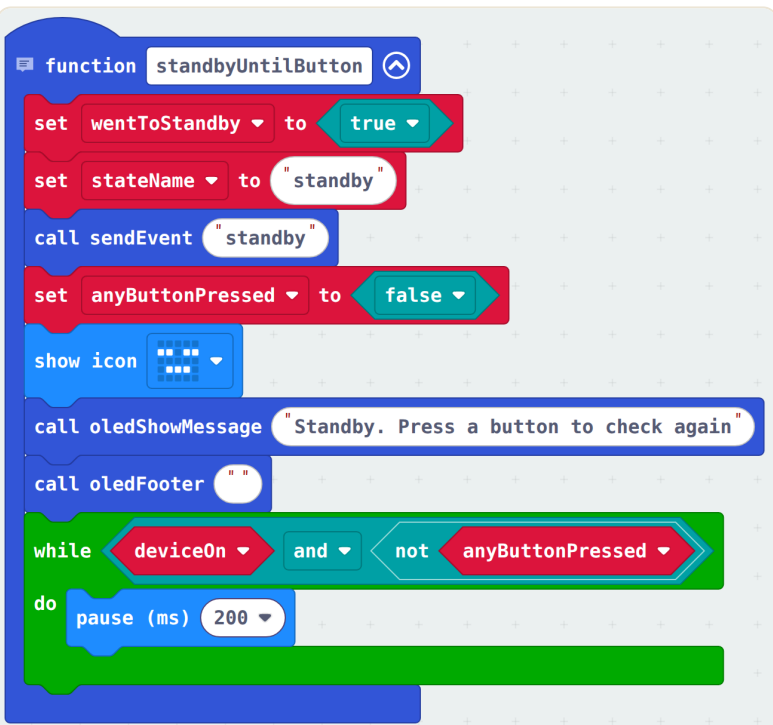
Used by: flashAndBeepUntilA, tapArmUntilA



tapArmUntilA A step of the program

Taps the wearer's arm with the motor until A is pressed. After 2 minutes it beeps instead, so a device that is not being worn does not run the motor flat.

Used by: extremeUv · Uses: flashAndBeepUntilA, stillWaitingForA, servoOff, servoTapOnce



standbyUntilButton A step of the program

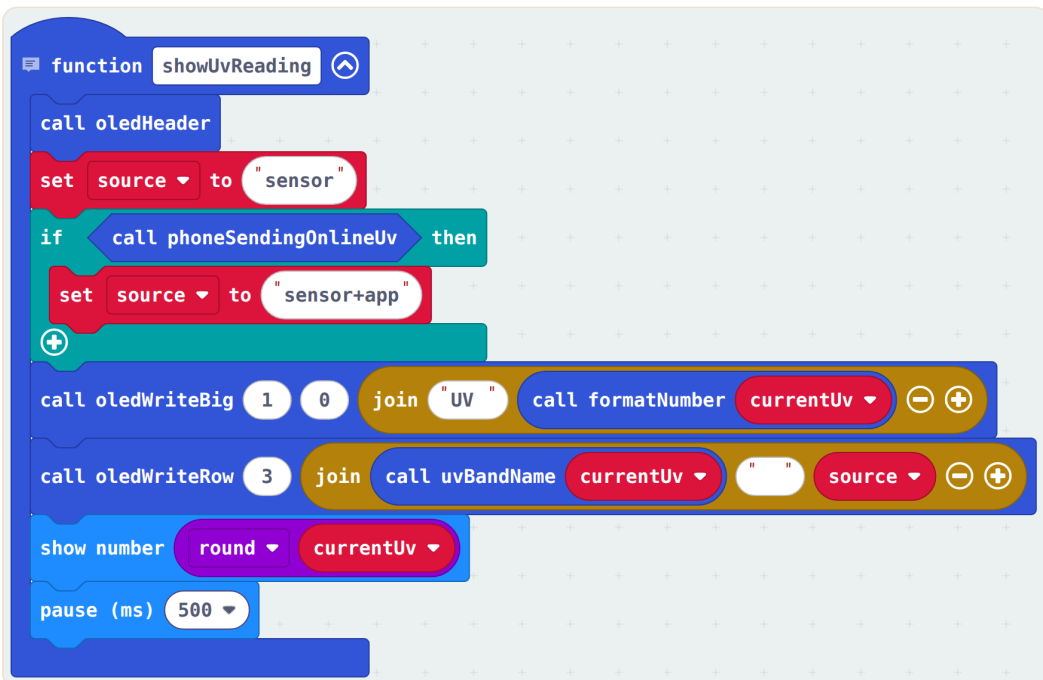
Sleep: waits for any button, or done from the app. Pressing A and B together still turns the device off. Afterwards it goes straight back to reading the sensor.

Used by: moderateOrHighUv, veryHighUv, extremeUv · Uses: oledShowMessage, oledFooter, sendEvent

5 Screen and lights

8 blocks

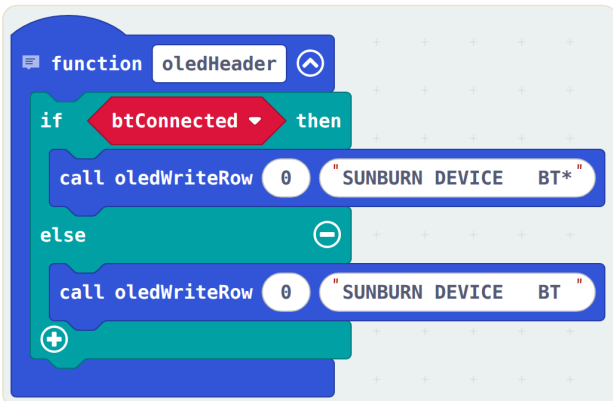
What you see. The 5 by 5 grid of lights shows the UV number, a happy face or a warning, and messages one word at a time. If a small screen is fitted, it shows the number large, the level, the message and the countdowns.



showUvReading A step of the program

Shows the UV number: rounded on the grid of lights, and exact, with the level and where it came from, on the screen.

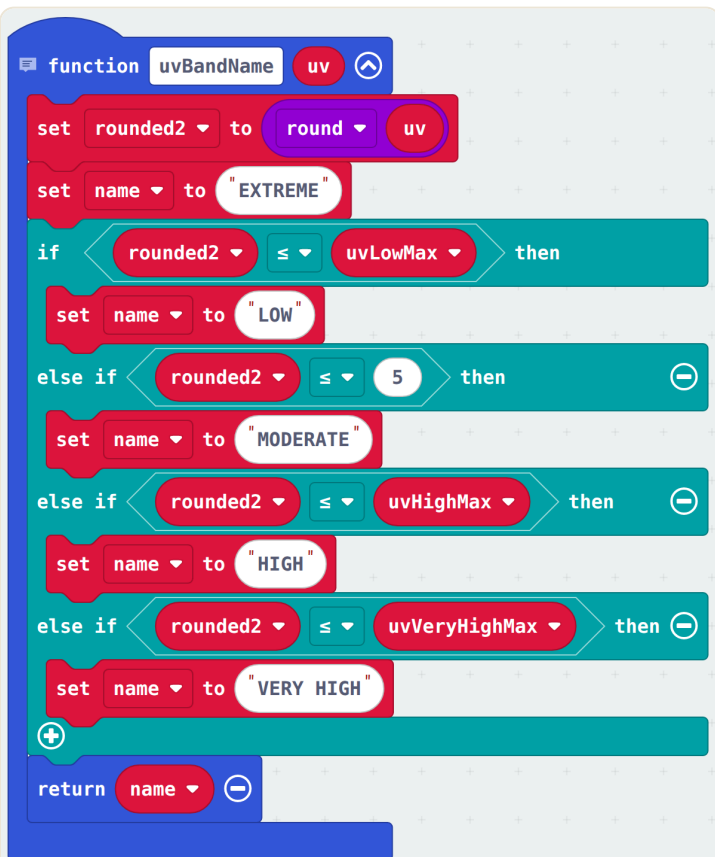
Used by: runFlowchart · Uses: phoneSendingOnlineUv, uvBandName, oledHeader, formatNumber, oledWriteRow, oledWriteBig



oledHeader A step of the program

The top line of the screen: the title, and a star while a phone is connected.

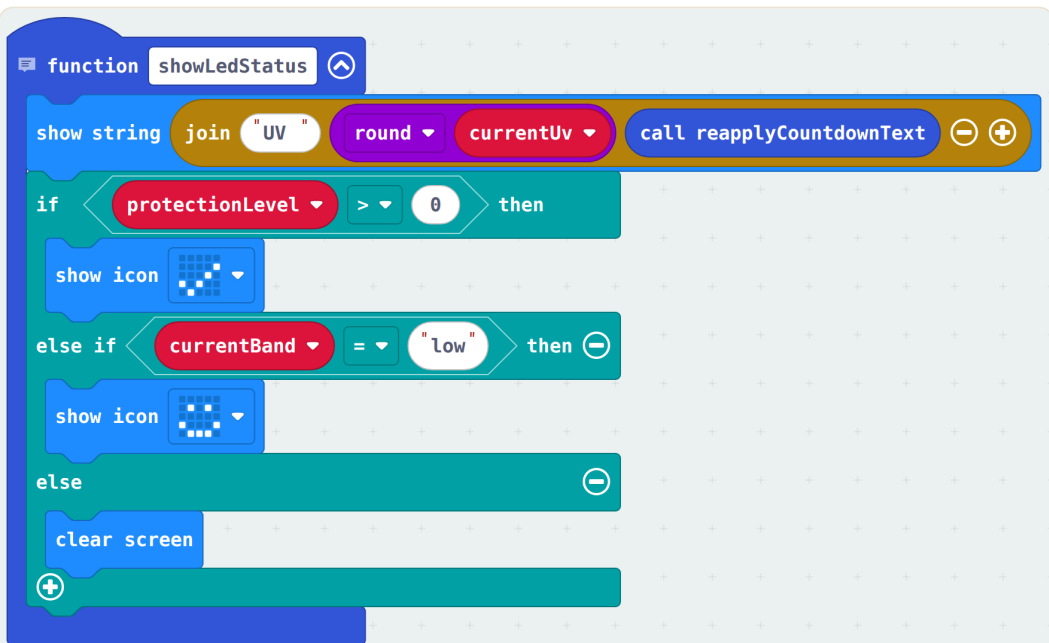
Used by: deviceTurnsOn, showUvReading, on bluetooth connected, on bluetooth disconnected · Uses: oledWriteRow



uvBandName Works out a piece of text · needs: uv

The words for the screen: LOW, MODERATE, HIGH, VERY HIGH or EXTREME.

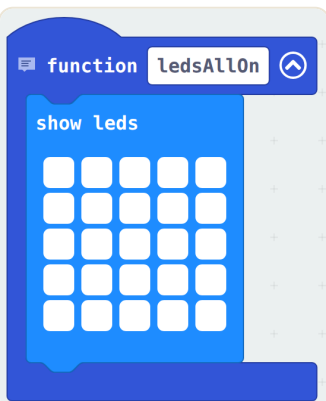
Used by: showUvReading



showLedStatus A step of the program

When A is pressed during a wait: the UV number and the time until reapply scroll across the lights, then the usual icon comes back.

Used by: wait5Minutes · Uses: reapplyCountdownText



ledsAllOn A step of the program

Turns all 25 lights on: the flash of an alert.

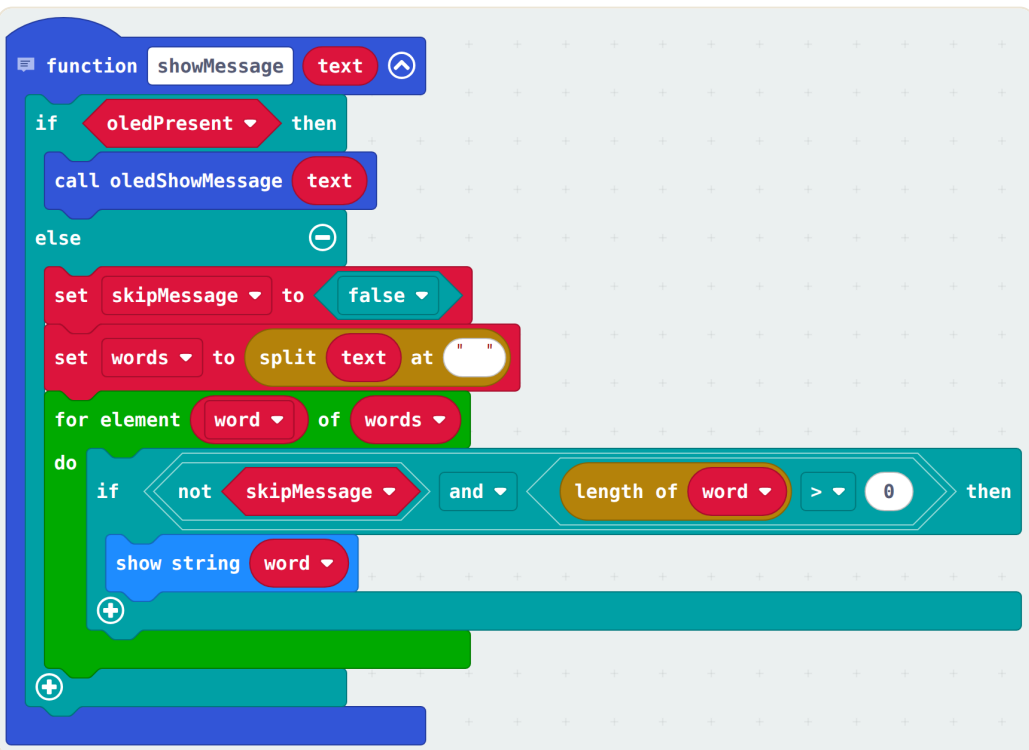
Used by: flashAndBeepUntilA



oledFooter A step of the program · needs: text

The bottom line of the screen: the countdowns.

Used by: showCheckSensor, wait5Minutes, deviceTurnsOn, standbyUntilButton · Uses: oledWriteRow



showMessage A step of the program · needs: text

Shows a message: on the screen if there is one, otherwise scrolled across the lights one word at a time. Pressing A skips the rest.

Used by: showCheckSensor, moderateOrHighUv, veryHighUv, extremeUv, showReapplySunscreen · Uses: oledShowMessage

5 Screen and lights (continued)

8 blocks

What you see. The 5 by 5 grid of lights shows the UV number, a happy face or a warning, and messages one word at a time. If a small screen is fitted, it shows the number large, the level, the message and the countdowns.

```
function oledShowMessage text
  set words2 to split text at ' '
  for wordIndex from 0 to length of array words2 - 1
    do
      set word2 to words2 get value at wordIndex
      if length of line == 0 then
        set line to word2
      else if length of line + 1 + length of word2 ≤ 21 then
        set line to join line ' ' word2
      else
        lines add value line to end
        set line to word2
      lines add value line to end
  for lineIndex from 0 to 2
    do
      if lineIndex < length of array lines then
        call oledWriteRow 4 + lineIndex lines get value at lineIndex
      else
        call oledWriteRow 4 + lineIndex ' '
```

oledShowMessage A step of the program · needs: text
The middle of the screen: a message, split over three lines of 21 letters.
Used by: showHappyFace, alreadyProtected, deviceTurnsOn, standbyUntilButton, showMessage · **Uses:** oledWriteRow

6 Sound and the tapping arm

5 blocks

The beeps, and the small motor (a servo) whose arm taps your wrist when the sun is extreme.

```
function setupSound
  if soundOutput == 2 then
    analog set pitch pin P0
    set volume soundVolume
```

setupSound A step of the program
Gets the sound ready: the micro:bit's built-in speaker, or a buzzer on pin P0, at the volume from the settings.
Used by: on start

```
function beep frequency ms
  play tone frequency for ms until done
```

beep A step of the program · needs: frequency, ms
One beep. It needs how high the beep should sound (the frequency) and how long it lasts (in milliseconds).
Used by: showCheckSensor, flashAndBeepUntilA

```
function servoAngle angle
  servo write pin P2 to angle
```

servoAngle A step of the program · needs: angle
Turns the motor's arm to an angle. The motor's signal wire is on pin P2.
Used by: servoOff, servoTapOnce

```
function servoOff
  if servoType == 2 then
    call servoAngle 90
  else
    call servoAngle servoRestAngle
  pause (ms) 300
  digital write pin P2 to 0
```

servoOff A step of the program
Moves the arm back to rest, then stops sending it signals so it is silent and uses no power.
Used by: deviceTurnsOn, deviceTurnsOff, tapArmUntilA · **Uses:** servoAngle

```
function servoTapOnce
  if servoType == 2 then
    call servoAngle 0
    pause (ms) 150
    call servoAngle 90
    pause (ms) 100
    call servoAngle 180
    pause (ms) 150
    call servoAngle 90
    pause (ms) 250
  else
    call servoAngle servoTapAngle
    pause (ms) 220
    call servoAngle servoRestAngle
    pause (ms) 330
```

servoTapOnce A step of the program
One tap of the arm, about half a second long.
Used by: tapArmUntilA · **Uses:** servoAngle

7 Talking to the phone

12 blocks

How the wearable and the phone app talk over Bluetooth. The phone sends the online UV index for where you are; the wearable sends back what it is doing every two seconds and reports each event, such as sunscreen applied. The same messages also go down the USB cable, so everything can be tested on a laptop. Every message ends with a checksum, so a message damaged on the way is thrown away instead of being misread.

```
function setupUsbLink
  serial redirect to USB
  serial set write line padding to 0
  if usbSpeed == 2 then
    serial set baud rate 9600
```

setupUsbLink A step of the program
Gets the USB link to a laptop ready; the same messages as Bluetooth go down the cable, without padding spaces, and at the slow speed if the settings ask for it.
Used by: on start

```
function advertiseOverBluetooth
  bluetooth uart service
```

advertiseOverBluetooth A step of the program
Makes the wearable visible over Bluetooth so a phone can connect.
Used by: on start

```
function sendEvent name
  call sendLine join 'ev=' name
```

sendEvent A step of the program · needs: name
Tells the app something happened: sunscreen applied, a reminder, went inside, sleep, a sensor problem, on or off.
Used by: extremeUv, sunscreenApplied, showReapplySunscreen, deviceTurnsOn, deviceTurnsOff, standbyUntilButton · **Uses:** sendLine

```
function calibrateToKnownUv knownUv
  call readUvSensor
  if knownUv > 0 and sensorVolts > uvZeroVolts then
    set uvVoltsPerIndex to sensorVolts - uvZeroVolts / knownUv
    call sendLine 'ok=cal'
  else
    call sendLine 'err=cal'
```

calibrateToKnownUv A step of the program · needs: knownUv
Corrects the sensor's scale. With the sensor in the sun, the app sends today's real UV index and the wearable adjusts its maths so it reads the same.
Used by: handleCommand · **Uses:** readUvSensor, sendLine

```
on bluetooth connected
  set btConnected to true
  call oledHeader
  call sendLine join 'hello;fw=' firmwareVersion
  call sendStatus
```

on bluetooth connected Runs when a phone connects
A phone has connected: say hello and send it the current status straight away.
Uses: oledHeader, sendLine, sendStatus

```
on bluetooth disconnected
  set btConnected to false
  call oledHeader
```

on bluetooth disconnected
Runs when the phone goes away
The phone has gone: remember that, and update the screen.
Uses: oledHeader

```
bluetooth on data received new line ( )
  call handleCommand bluetooth uart read until new line ( )
```

on bluetooth data received Runs when a line of text arrives from the phone
A line of text arrived from the phone: pass it to the block that reads commands.
Uses: handleCommand

```
every 2000 ms
  call sendStatus
```

every 2000 ms Runs every 2 seconds in the background
Every 2 seconds, send the app a status line.
Uses: sendStatus

```
serial on data received new line ( )
  call handleCommand serial read until new line ( )
```

on serial data received Runs when a line of text arrives over the USB cable
A line of text arrived over the USB cable: pass it to the block that reads commands.
Uses: handleCommand

```
function sendLine text
  set line2 to join text 'ck=' call lineChecksum text
  serial write line line2
  if btConnected then
    bluetooth uart write line line2
```

sendLine A step of the program · needs: text
Sends one message, with its checksum on the end, down the USB cable and to the phone when one is connected.
Used by: showCheckSensor, on bluetooth connected, sendEvent, sendStatus, handleCommand, calibrateToKnownUv · **Uses:** lineChecksum

```
function sendStatus
  set reapplySeconds to -1
  if reapplyTimerRunning then
    set msLeft to reapplyDueTime - running time (ms)
    set reapplySeconds to max of 0 and floor msLeft / 1000
  set onlineText to '-1'
  if call phoneSendingOnlineUv then
    set onlineText to call formatNumber onlineUv
  set status to join 'st=' stateName
  set status to join status ';uv=' call formatNumber currentUv
  set status to join status ';sen=' call formatNumber max of 0 and sensorUv
  set status to join status ';onl=' onlineText
  set status to join status ';band=' call uvBand currentUv
  set status to join status ';spf=' reapplySeconds
  set status to join status ';spfnt=' sunscreenCount
  set status to join status ';fw=' firmwareVersion
  if demoMode then
    set status to join status ';demo=1'
  else
    set status to join status ';demo=0'
  set status to join status ';pvt=' call formatNumber sensorVolts
  call sendLine status
```

sendStatus A step of the program
Builds the status line the app shows: what the device is doing, the UV values, the level, the seconds until sunscreen is due, how many applications so far, the program version, whether demo timings are on, and the voltage on the sensor pin.
Used by: on bluetooth connected, every 2000 ms, handleCommand · **Uses:** phoneSendingOnlineUv, uvBand, sendLine, formatNumber

7

Talking to the phone (continued)

12 blocks

How the wearable and the phone app talk over Bluetooth. The phone sends the online UV index for where you are; the wearable sends back what it is doing every two seconds and reports each event, such as sunscreen applied. The same messages also go down the USB cable, so everything can be tested on a laptop. Every message ends with a checksum, so a message damaged on the way is thrown away instead of being misread.

```
function handleCommand command
  set parts to split call cleanText command at '-'
  set key to parts get value at 0
  if length of array parts > 1 then
    set value to parts get value at 1
  +
  call debugLog join 'cmd' key ' ' value
  if key == 'uv' then
    set newUv to parse to number value
    if newUv >= 0 and newUv <= 20 then
      set onlineUv to newUv
      set onlineUvTime to running time (ms)
      call sendLine 'ok=uv'
    else
      call sendLine 'err=uv'
  +
  else if key == 'ack' then
    set ackPressed to true
    set anyButtonPressed to true
    call sendLine 'ok=ack'
  else if key == 'zero' then
    call readUvSensor
    set uvZeroVolts to sensorVolts
    call sendLine join 'ok=zero;v=' call formatNumber uvZeroVolts
  else if key == 'cal' then
    call calibrateToKnownUv parse to number value
  else if key == 'demo' then
    set demoMode to value == 1
    call applyTimings
    call sendLine 'ok=demo'
  else if key == 'power' then
    if value == 0 and deviceOn then
      call deviceTurnsOff
    else if value == 1 and not deviceOn then
      call deviceTurnsOn
  +
  else if key == 'debug' then
    set debugMode to value == 1
    call sendLine 'ok=debug'
  else if key == 'vol' then
    set soundVolume to parse to number value
    set volume soundVolume
    call sendLine 'ok=vol'
  else if key == 'ping' then
    set oledText to 0
    if oledPresent then
      set oledText to 1
    +
    call sendLine join 'pong;fw=' firmwareVersion 'oled=' oledText
  else if key == 'read' then
    call sendStatus
  else if length of key > 0 then
    call sendLine 'err=unknown'
```

handleCommand A step of the program · needs: command
Reads a command from the app or a laptop and acts on it: a new online UV value, done, calibrate the sensor, demo timings on or off, switch off or on, extra detail on or off, the beep volume, ping, or send the status now.
Used by: on bluetooth data received, on serial data received · **Uses:** readUvSensor, deviceTurnsOn, deviceTurnsOff, sendLine, sendStatus, calibrateToKnownUv, applyTimings, formatNumber, cleanText, debugLog

8

Helpers

11 blocks

Small jobs the other blocks share: the maths for each type of sensor, turning minutes into the milliseconds the program counts in, sorting numbers, tidying up numbers, times and text, and the checksum that goes on the end of every message.

```
function applySensorPreset
  set uvZeroVolts to 0
  set uvVoltsPerIndex to 0.1
  if uvSensorType == 2 then
    set uvZeroVolts to 1
    set uvVoltsPerIndex to 0.12
  +
  if uvSensorType == 3 then
    set uvVoltsPerIndex to 0.22
```

applySensorPreset A step of the program
The maths for each type of UV sensor: which voltage means zero, and how many volts make one point of UV index.
Used by: on start

```
function twoDigits n
  set digits to join n
  if n < 10 then
    set digits to join '0' n
  +
  return digits
```

twoDigits Works out a piece of text · needs: n
Adds a leading zero: 7 becomes 07.
Used by: formatTime

```
function isBlank code
  return code == 32 or code == 13 or code == 10
```

isBlank A yes-or-no question the program asks · needs: code
Asks: is this character a space or the end of a line?
Used by: cleanText

```
function applyTimings
  if demoMode then
    set waitMs to 10000
    set reapplyMs to 60000
    set alertGiveUpMs to 20000
    set tapMaxMs to 15000
  else
    set waitMs to waitMinutes * 60000
    set reapplyMs to reapplyHours * 3600000
    set alertGiveUpMs to alertGiveUpMinutes * 60000
    set tapMaxMs to tapMaxMinutes * 60000
```

applyTimings A step of the program
Turns the minutes and hours from the settings into the milliseconds the program counts in. Demo timings are much shorter, so the whole flowchart can be shown in a few minutes.
Used by: handleCommand, on start

```
function sortNumbers list
  for sortIndex from 0 to length of array list - 1 - 1
  do
    set movingValue to list get value at sortIndex + 1
    set slot to sortIndex
    while slot >= 0 and list get value at slot > movingValue
    do
      list set value at slot + 1 to list get value at slot
      set slot to slot - 1
    list set value at slot + 1 to movingValue
```

sortNumbers A step of the program · needs: list
Puts a list of numbers in order from smallest to largest. It is used to find the middle sensor reading.
Used by: readUvSensor

```
function reapplyCountdownText
  if reapplyTimerRunning then
    set timeLeft3 to reapplyDueTime - running time (ms)
    set countdown to join 'SPF ' call formatTime timeLeft3
  +
  return countdown
```

reapplyCountdownText Works out a piece of text
The short SPF 1h23 text for the bottom of the screen while the sunscreen timer is running.
Used by: wait5Minutes, showLedStatus · **Uses:** formatTime

```
function debugLog text
  if debugMode then
    serial write line join 'dbg:' text
```

debugLog A step of the program · needs: text
Extra detail lines for a laptop, sent only after the command debug=1.
Used by: readUvSensor, handleCommand

```
function formatNumber n
  set rounded3 to round n * 10 / 10
  set wholePart to floor absolute of rounded3
  set tenthPart to round absolute of rounded3 - wholePart * 10
  if rounded3 < 0 then
    set sign to '-'
  +
  return join sign wholePart '.' tenthPart
```

formatNumber Works out a piece of text · needs: n
Writes a number with one decimal place: 7.25 becomes 7.3.
Used by: showCheckSensor, showUvReading, sendStatus, handleCommand

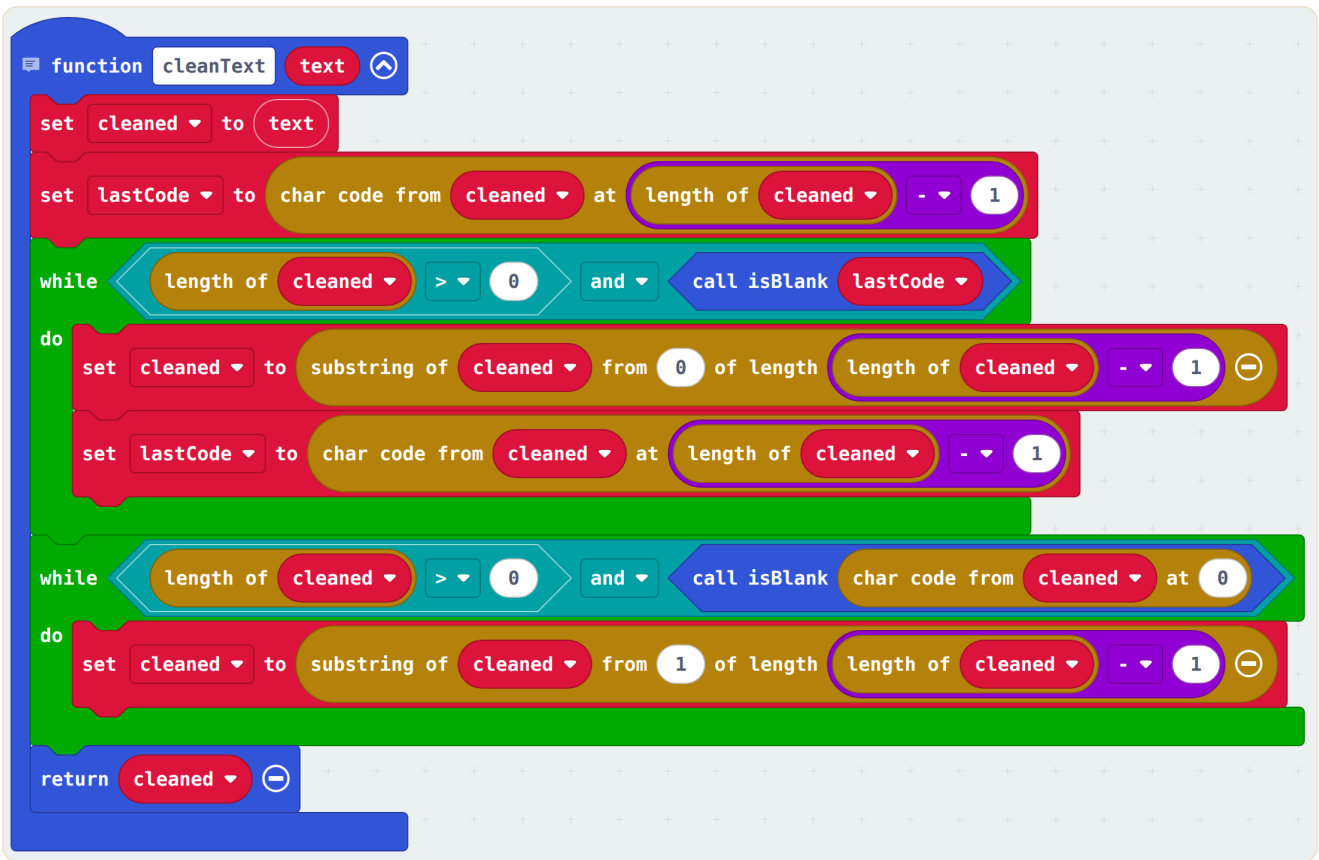
```
function formatTime ms
  set totalSeconds to floor max of 0 and ms / 1000
  set hours to floor totalSeconds / 3600
  set minutes to remainder of floor totalSeconds / 60 / 60
  set seconds to remainder of totalSeconds / 60
  if hours == 0 then
    set timeText to join minutes ' ' call twoDigits seconds
  else
    set timeText to join hours 'h' call twoDigits minutes
  +
  return timeText
```

formatTime Works out a piece of text · needs: ms
Writes a time the way a clock would: 1:30, or 2h00.
Used by: alreadyProtected, wait5Minutes, reapplyCountdownText · **Uses:** twoDigits

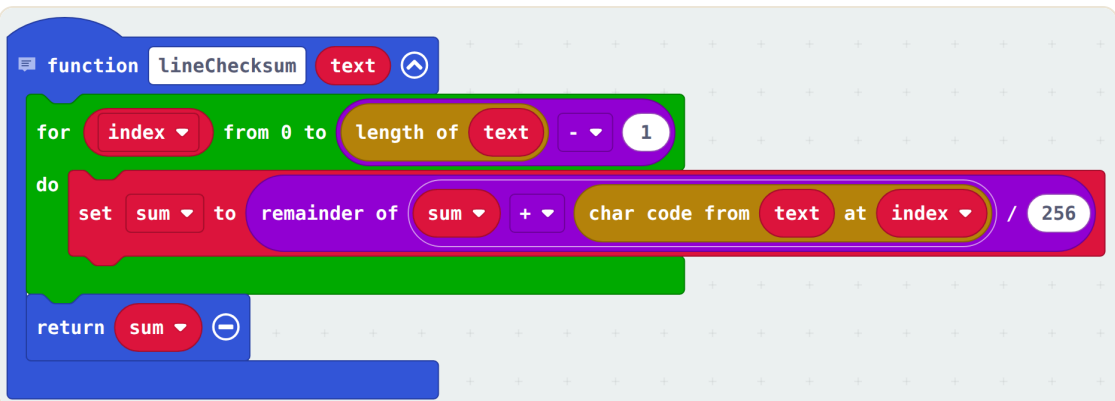
8 Helpers (continued)

11 blocks

Small jobs the other blocks share: the maths for each type of sensor, turning minutes into the milliseconds the program counts in, sorting numbers, tidying up numbers, times and text, and the checksum that goes on the end of every message.



cleanText Works out a piece of text · needs: text
Removes stray spaces and line endings from a command before it is read.
Used by: handleCommand · **Uses:** isBlank

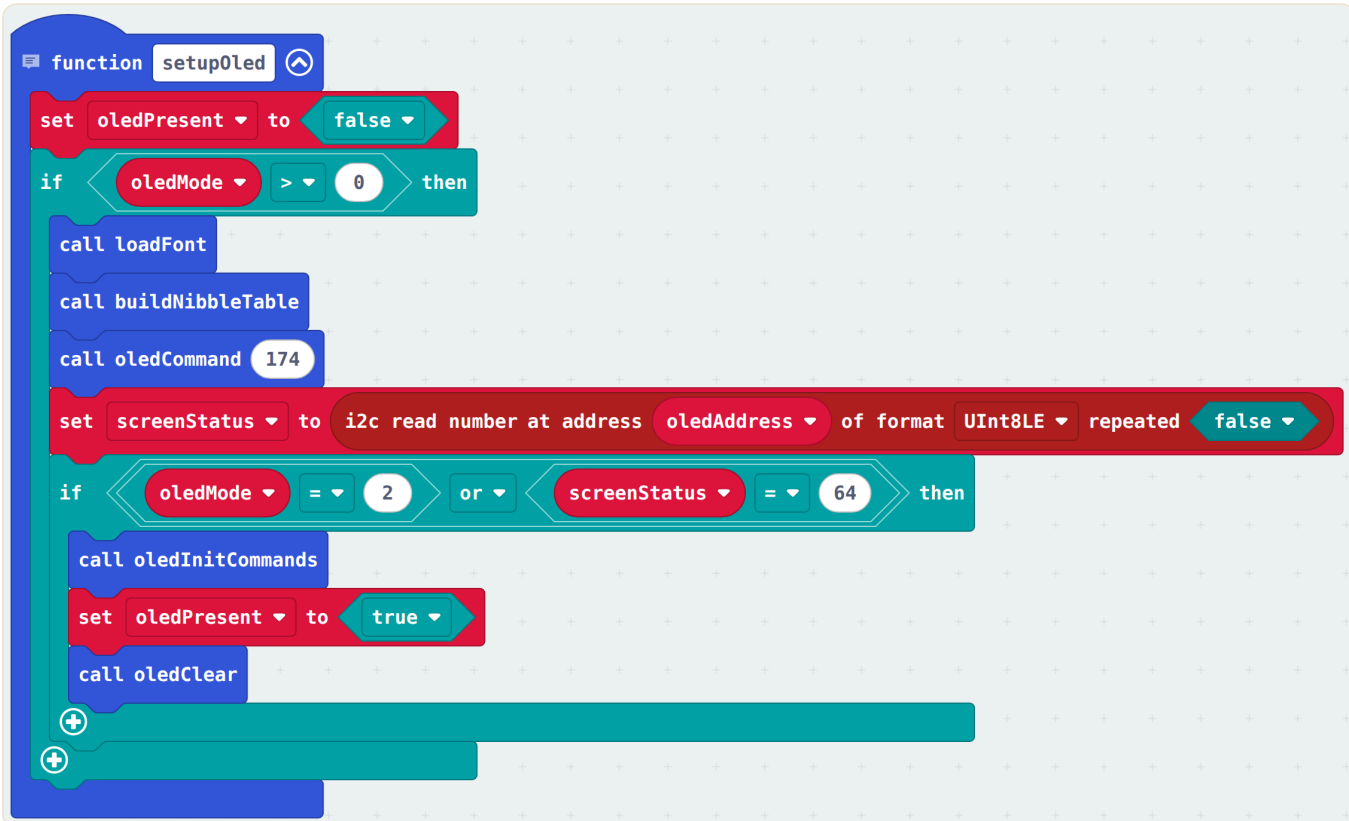


lineChecksum Works out a number · needs: text
Adds up the character codes of a message (0 to 255). It goes on the end of every message, so the app can tell when a message was damaged on the way and ignore it.
Used by: sendLine

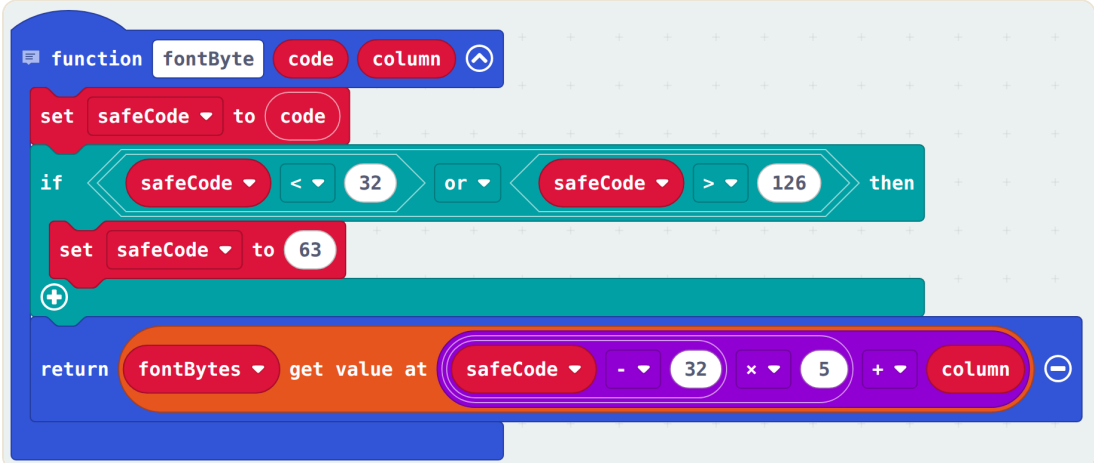
9 The optional screen

12 blocks

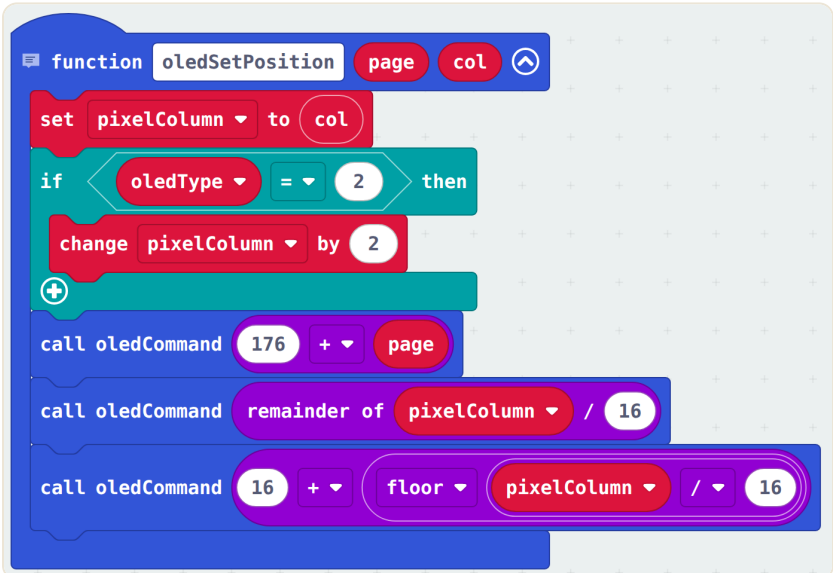
Only used if a small screen is added to the wearable. These blocks draw letters and numbers on it dot by dot. The device works exactly the same without a screen.



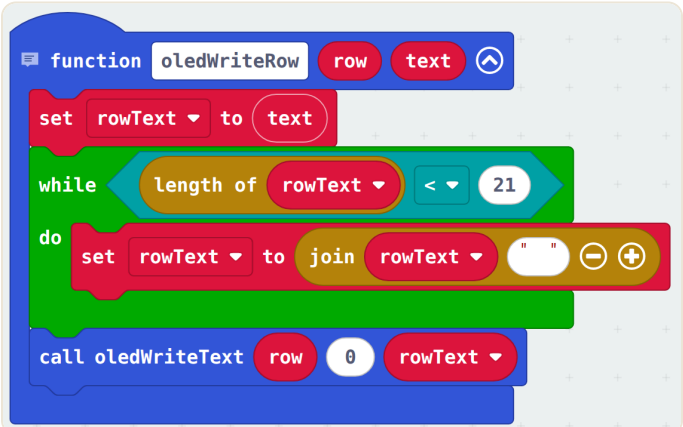
setupOled A step of the program
Looks for a screen on pins 19 and 20 and gets it ready. If there is no screen, everything else works the same.
Used by: on start · **Uses:** oledInitCommands, loadFont, buildNibbleTable, oledCommand, oledClear



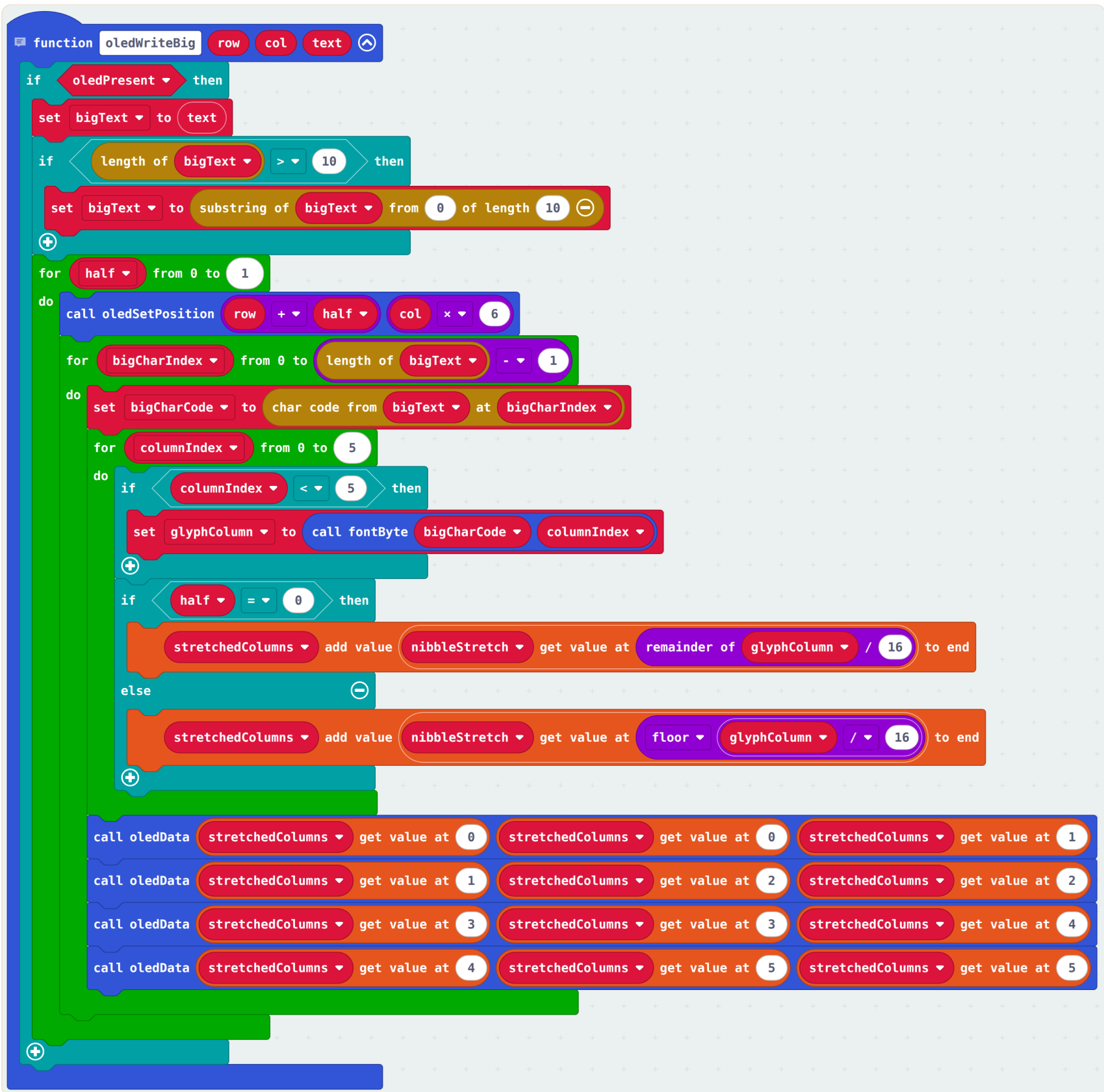
fontByte Works out a number · needs: code, column
One column of dots, out of five, of one character.
Used by: oledWriteText, oledWriteBig



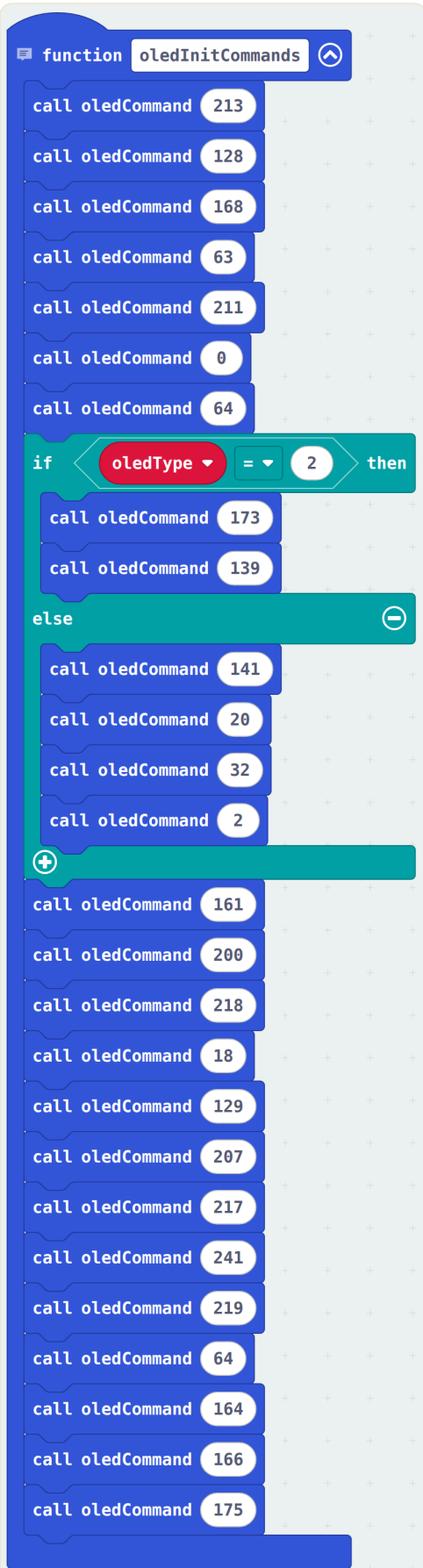
oledSetPosition A step of the program · needs: page, col
Moves to a place on the screen: a row from 0 to 7 and a column from 0 to 127.
Used by: oledClear, oledWriteText, oledWriteBig · **Uses:** oledCommand



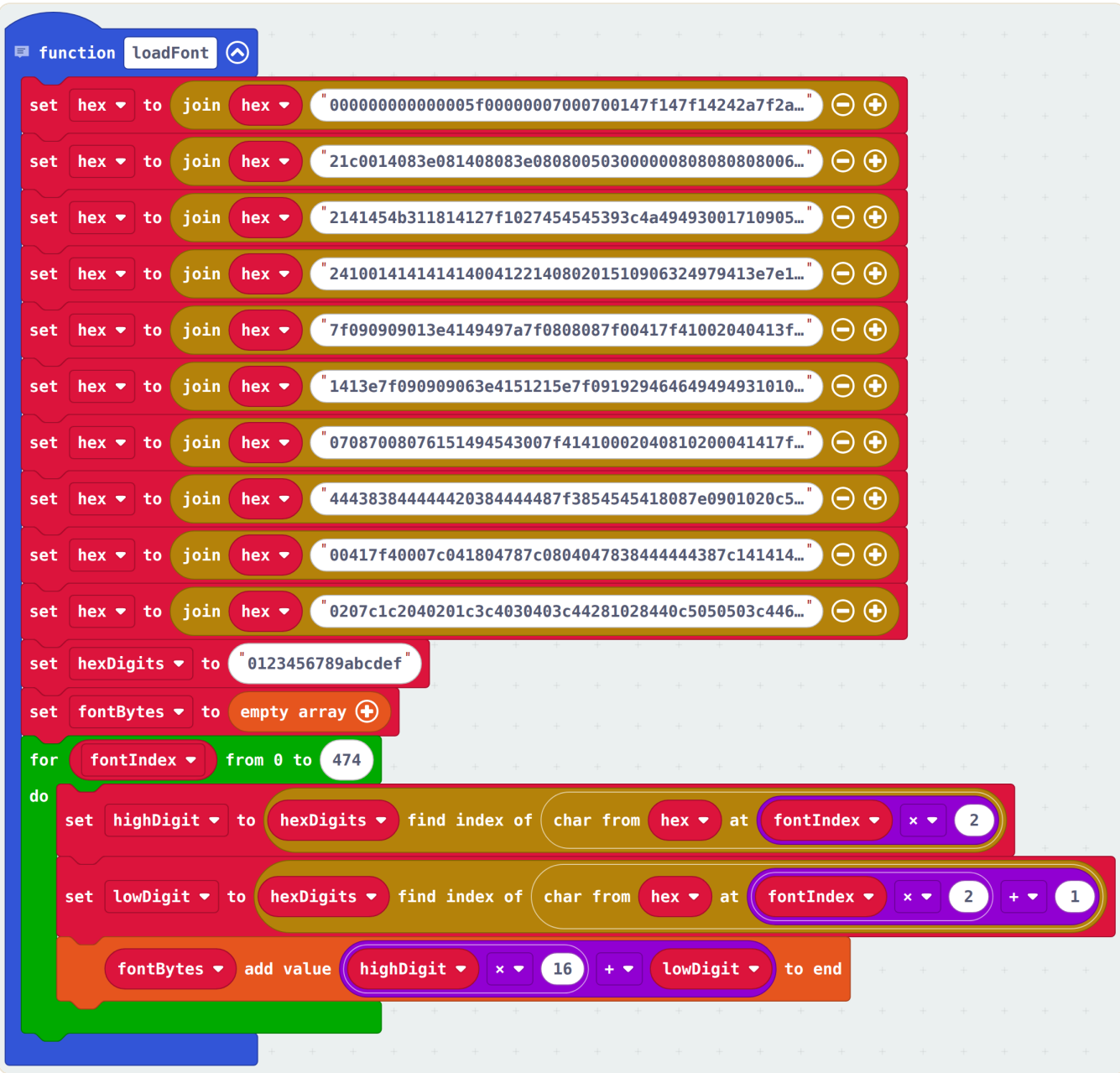
oledWriteRow A step of the program · needs: row, text
Writes a whole row, filling the rest with spaces so old text is wiped.
Used by: deviceTurnsOff, showUvReading, oledHeader, oledShowMessage, oledFooter · **Uses:** oledWriteText



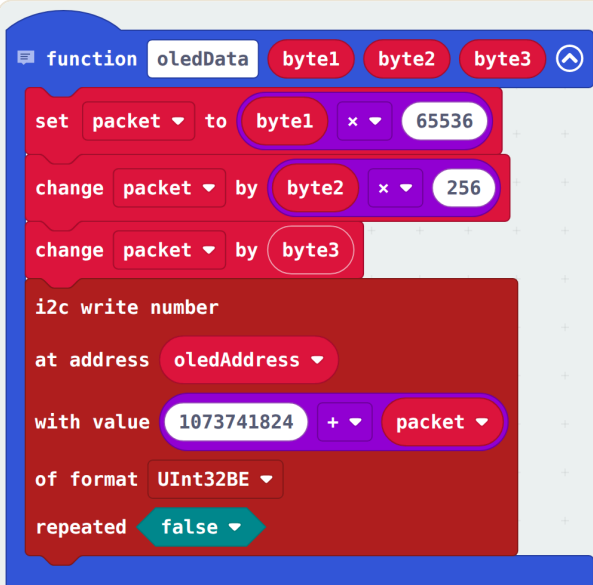
oledWriteBig A step of the program · needs: row, col, text
Writes double-size text, two rows tall and up to 10 characters. It is used for the big UV number.
Used by: showUvReading · **Uses:** fontByte, oledData, oledSetPosition



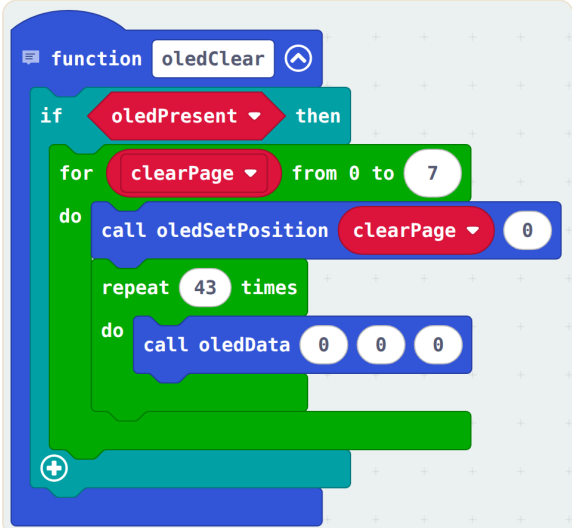
oledInitCommands A step of the program
The standard list of instructions that wakes up a 128 by 64 dot screen.
Used by: setupOled · **Uses:** oledCommand



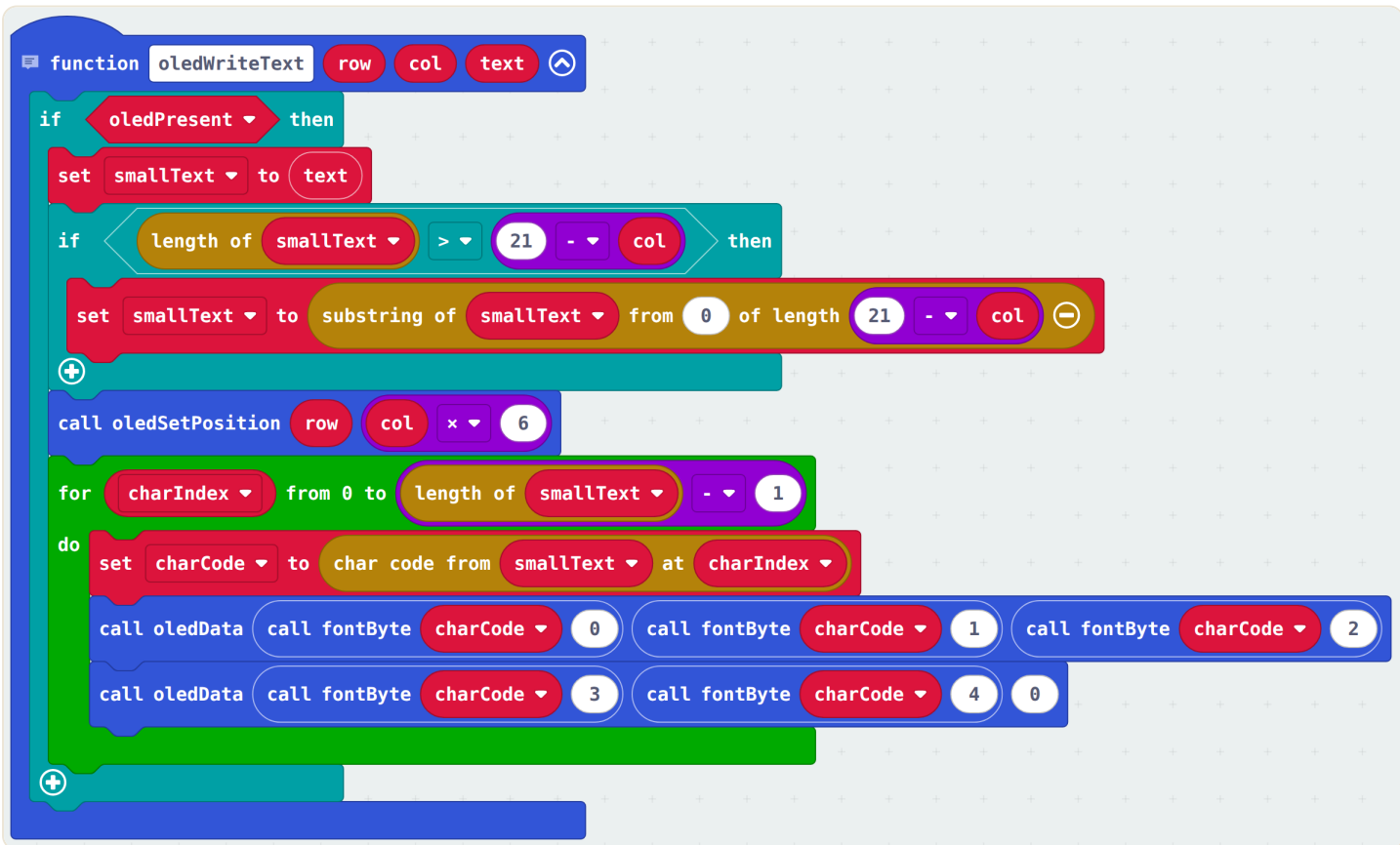
loadFont A step of the program
The shape of every letter, number and symbol, 5 dots wide by 7 tall, stored as text and unpacked when the device starts.
Used by: setupOled



oledData A step of the program · needs: byte1, byte2, byte3
Sends the screen three columns of dots.
Used by: oledClear, oledWriteText, oledWriteBig



oledClear A step of the program
Blanks the whole screen.
Used by: deviceTurnsOff, setupOled · **Uses:** oledData, oledSetPosition



oledWriteText A step of the program · needs: row, col, text
Writes small text: up to 21 characters on each of 8 rows.
Used by: oledWriteRow · **Uses:** fontByte, oledData, oledSetPosition